What Data Science Is

Data science is a disciplined way to learn from data. It brings together subject-area knowledge, programming, statistical reasoning, and communication. The goal is not simply to produce charts or run algorithms; it is to create reliable evidence that helps answer a clearly stated question.

An analysis begins by understanding a real problem and the people affected by it. The analyst then identifies what evidence is needed, where suitable data may come from, and what limits apply. Results should be explained in language that decision-makers and the public can understand.

Key points

- Data are observations collected for a purpose.
- Analytics describes and investigates patterns; machine learning learns patterns for prediction or decision support.
- A useful result is relevant, reproducible, understandable, and responsibly produced.

Data Types and Measurement

A dataset is commonly arranged as observations and variables. An observation is the entity or event described by a row; a variable records one characteristic in a column. Correctly identifying the observation unit prevents mistakes such as counting the same person several times or mixing measurements from different time periods.

Variables may be numeric, categorical, text, date/time, or identifiers. Measurement scales also matter: categories may be nominal or ordered, while numeric values may be counts or measurements. A postal code may look numeric but should usually be treated as a category, not averaged.

Key points

- Structured data use a defined table or schema; semi-structured data include JSON; unstructured data include images and free text.
- Check units, allowed values, ranges, and whether a field is an identifier.
- Document the source, collection date, and meaning of important fields.

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The Data Science Lifecycle

A typical workflow moves through problem framing, data acquisition, preparation, exploration, analysis or modeling, evaluation, and communication. These are not rigid one-way steps. Exploration may reveal that the original question is unclear, while evaluation may show that more appropriate data are required.

Problem framing turns a broad concern into an answerable question. A good question identifies the population, outcome, time period, and comparison of interest. It also defines what a useful answer would look like and what decisions the analysis can reasonably inform.

Key points

- Ask: define the question, audience, and success criteria.
- Prepare and explore: assess quality, distributions, and limitations.
- Evaluate and communicate: test whether evidence supports the claim and explain uncertainty.

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Tools and Reproducible Work

Python is widely used for data workflows. Notebooks combine code, output, and explanatory text, while scripts package repeatable tasks. Libraries such as NumPy and pandas support numerical and tabular work; Matplotlib and Seaborn support visualization. Tool choice should follow the task rather than determine it.

Reproducibility means another person can follow the steps and understand how the results were made. Keep raw data unchanged, record transformations, use clear file names, and state software assumptions. A notebook should run from its first cell to its last without depending on hidden state.

Key points

- Separate original data from processed outputs.
- Use descriptive variable names and small, testable steps.
- Record software versions, sources, assumptions, and any manual decisions.

Responsible Data Science

Data work can affect privacy, access, and opportunity. Before collecting or reusing data, check permission, consent, terms of use, and whether sensitive information is necessary. Remove or protect personal identifiers where appropriate, and do not publish confidential records or access credentials.

Bias can enter through who is represented, how variables are measured, and how outcomes are interpreted. A model or summary may perform differently across groups. Analysts should examine who may be missing, who may be harmed, and whether an apparently neutral measure encodes unequal conditions.

Key points

- Use only data you are permitted to access and use.
- Minimize personal data and protect files and credentials.
- Describe limitations and avoid claims that the evidence cannot support.

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Guided Study and Mini-Project

Choose a small, openly available dataset and identify its source, owner, date, license, and intended use. Write down the observation unit and explain the most important variables. Check the number of rows, missing fields, duplicates, and possible sensitive information before attempting analysis.

Frame one measurable question and list the columns needed to answer it. Prepare a short data profile, then write a finding that refers to specific evidence. State what the dataset cannot establish—for example, a pattern in one sample may not describe every population or prove a cause.

Key points

- Deliver: question, source and provenance, observation unit, variables, and data-quality notes.
- Include one evidence-based finding and one limitation.
- Review: What distinguishes an observation from a variable? Why should an analysis begin with a question?